

Molecular Orbital Theory

Theory was proposed by Hund & Mulliken in 1932. Molecular Orbital theory is a quantum mechanical approach that describes the distribution of electrons within molecules. It explains how atomic orbitals combine to form molecular orbitals, which define electronic structure of a molecule.

Core Principles:-

- 1- Linear combination of atomic orbitals:-
Molecular orbitals are formed by linear combination of atomic orbitals.
- 2- Self-consistent fields:-
The molecular orbital coefficients are optimized to minimize the energy of the system.
- 3- Variational Principles:-
The molecular orbital wave function is varied to minimize the energy.
- 4- Schrodinger Equations:-
The molecular orbital wave function satisfies the time-independent Schrodinger Equation.

Postulates of MOT:-

According to MOT:-

- 1- The molecular orbital approach considers the whole molecule as a single unit.
- 2- The atomic orbitals of combining atoms overlap to form new orbitals called "Molecular Orbitals".
- 3- The molecular orbitals surround two or more nuclei of the bonded atoms.
- 4- Two atomic orbitals after overlapping form two molecular orbitals which differ in energy.

=> Bonding Molecular Orbital:-

The molecular orbitals that is obtained by addition overlap of two atomic orbitals is termed as Bonding Molecular Orbital.

$$\psi^b = \psi^A + \psi^B$$

- => BMO possesses lower energy than the atomic orbitals
- => BMO has high electron density

=> Anti Bonding Molecular Orbitals:-

The molecular orbital obtained by the subtraction of overlap of two atomic orbitals is termed as



Anti bonding molecular orbitals.

$$\psi^* = \psi_A - \psi_B$$

- $\Rightarrow$ ABMO possesses higher energy
 $\Rightarrow$ The probability of finding electron
 in between the nuclei is negligible.

5- "n" atomic orbitals combine to form "n" molecular orbitals.

6- The filling of electrons into the molecular orbital takes

place according to:-

- 1- Auf bau principle
- 2- Pauli's exclusion principle
- 3- Hund's rule

Combination of s orbital to give Molecular orbitals:-

Consider two hydrogen atoms with their s-orbitals. They have 1 electron each.

The first combination is obtained by addition of two atomic orbitals. The pictorial representation is shown.

$$\psi^{MO} = N(\psi_{1s_a} + \psi_{1s_b})$$

where N is normalizing constant

The electron in this MO will spend most of its time between

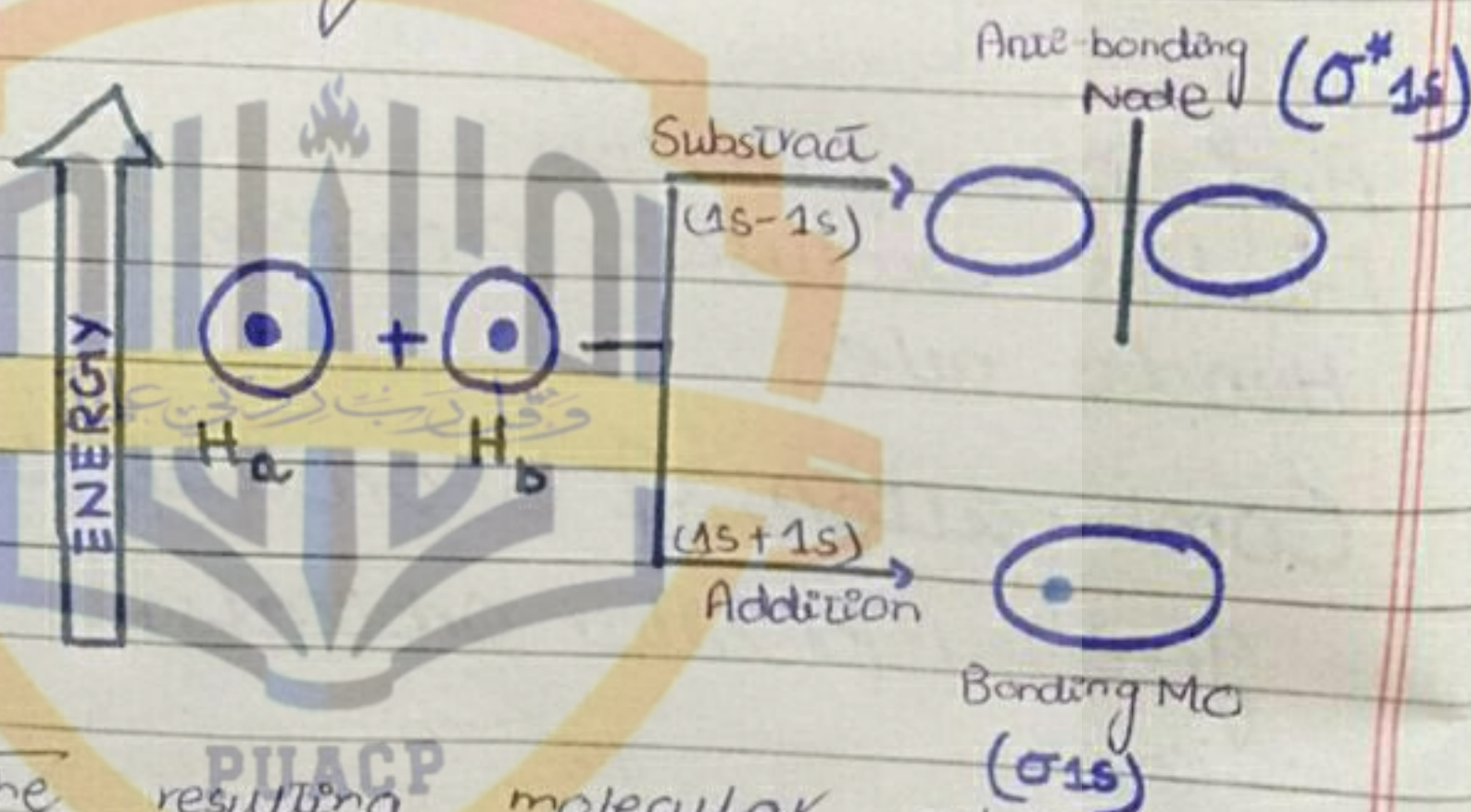
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The two nuclei H_a and H_b .

The electron is now stabilized by two nuclear charges instead of one in an atomic orbital. The resulting MO is of lower energy than 1s orbital. That is why it is called a **BMO**.

The second possibility of linear combination is obtained by subtracting the two atomic orbitals.



The resulting molecular orbital has zero electron density between the nuclei. The region of zero electron density is called nodal plane.

Bond Order from Number

of Electrons in BMO & ABMO

⇒ If the number of electrons in BMO are greater than the number of electrons in ABMO's then the molecule would be stable.

⇒ If the number of electrons in BMO's are equal to the no. of electrons in ABMO's then the molecule is unstable and it does not exist.

Bond Order:-

It is defined as the number of covalent bonds present in a molecule.

$$\text{Bond Order} = \frac{\text{No. of } e^- \text{ in BMO} - \text{No. of } e^- \text{ in ABMO}}{2}$$

Molecular Energy level Diagram for H_2 molecule:-

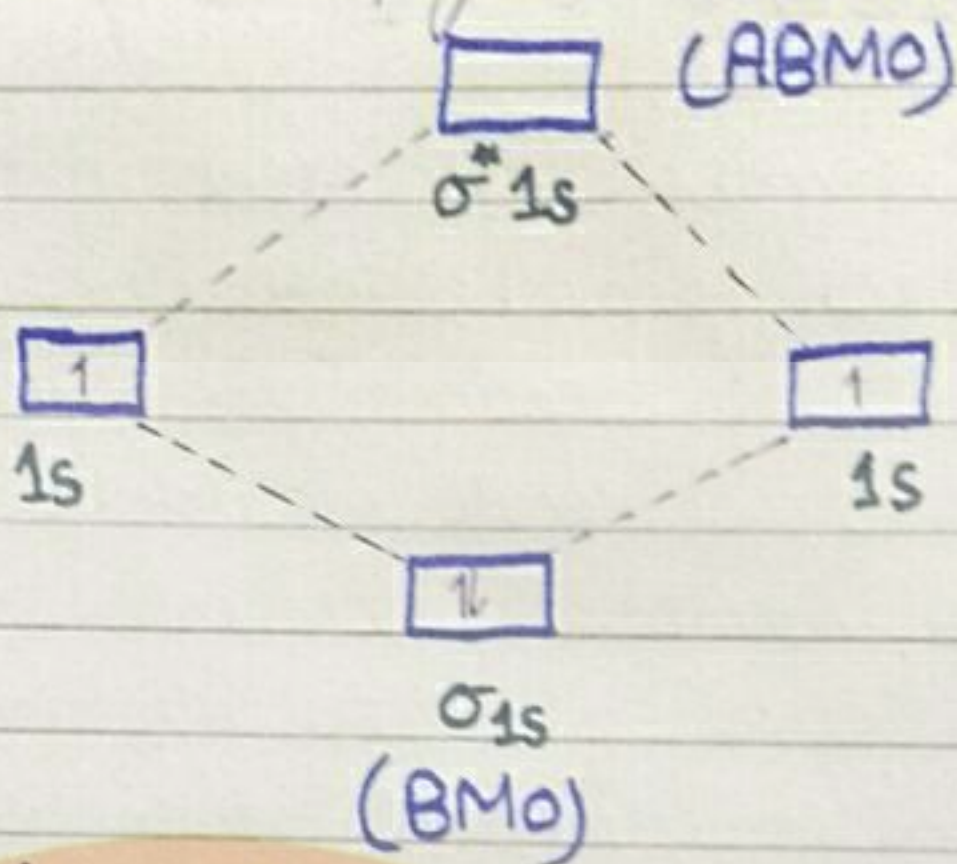
H_2 is formed by overlap of 1s atomic orbitals. It shall have two molecular orbitals.

One is bonding σ_{1s} and one is

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Day: _____

Anti bonding $\sigma^* 1s$.
The energy level diagram of H_2 is shown in fig.



Bond order:-

$$H_2 = \frac{2 - 0}{2} = 1$$

Bond energy:-

Bond dissociation energy in H_2 is $431.4 \text{ kJ mol}^{-1}$

Bond length:-

Bond length is 74 pm .

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Molecular Orbital Energy level for H_2 Molecule:-

The electronic configuration of He is $1s^2$.

The $1s$ orbital of He atoms



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combine to form one bonding $\sigma(1s)$ and one anti-bonding $\sigma^*(1s)$. Total four electrons are involved. Two electrons would go in the bonding σ_{1s} MO and the other two would have to go into the anti-bonding σ_{1s}^* MO.

Bond order:-

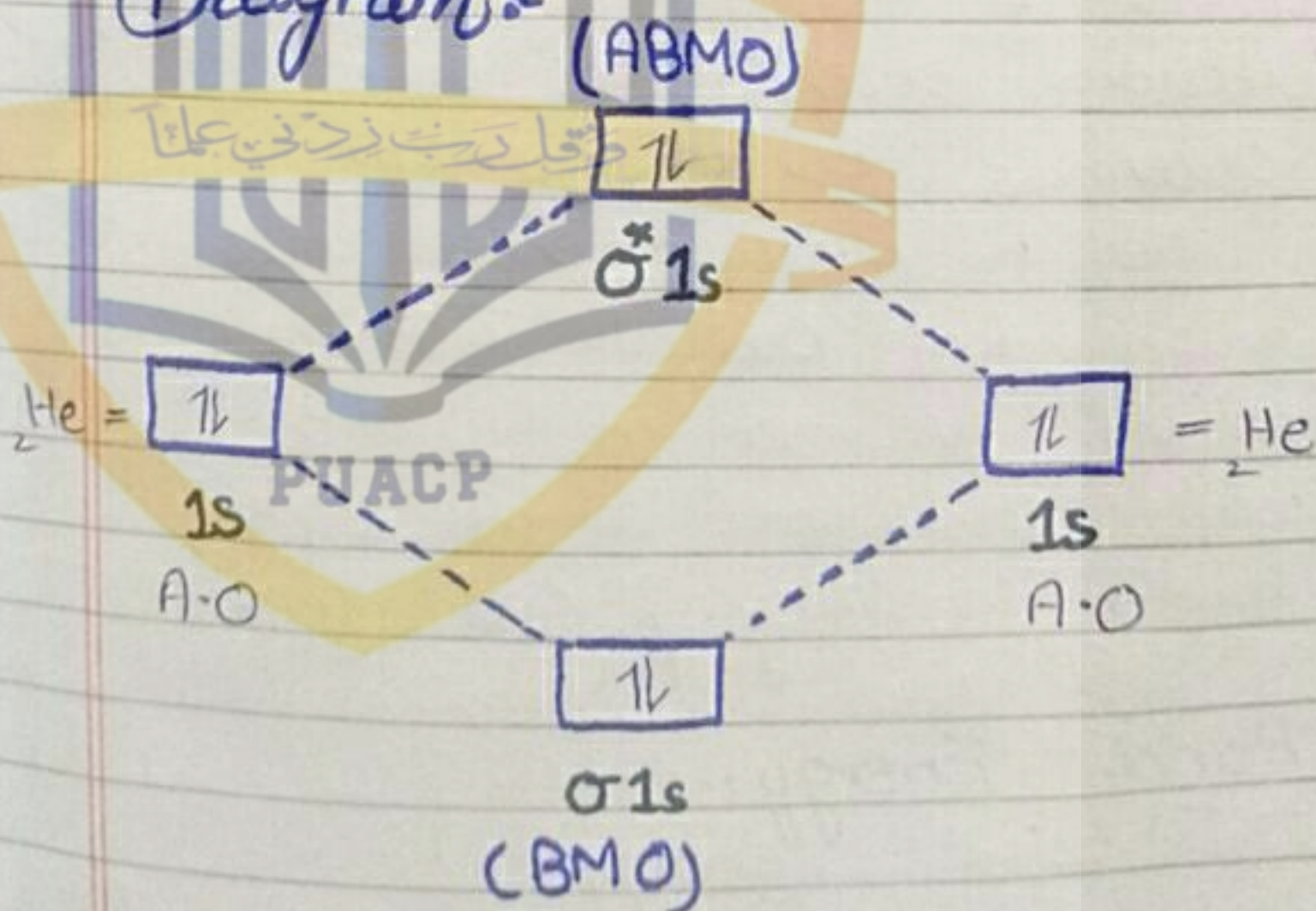
The bond order of He_2 is zero

$$= \frac{2 - 2}{2} = \frac{0}{2} = 0$$

Bond energy:-

Bond dissociation of Helium is 0.0214 eV or 0.0004 eV.

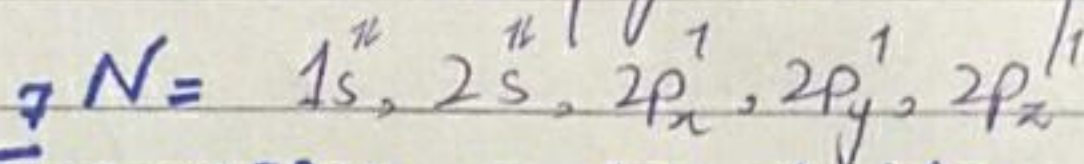
Diagram:-



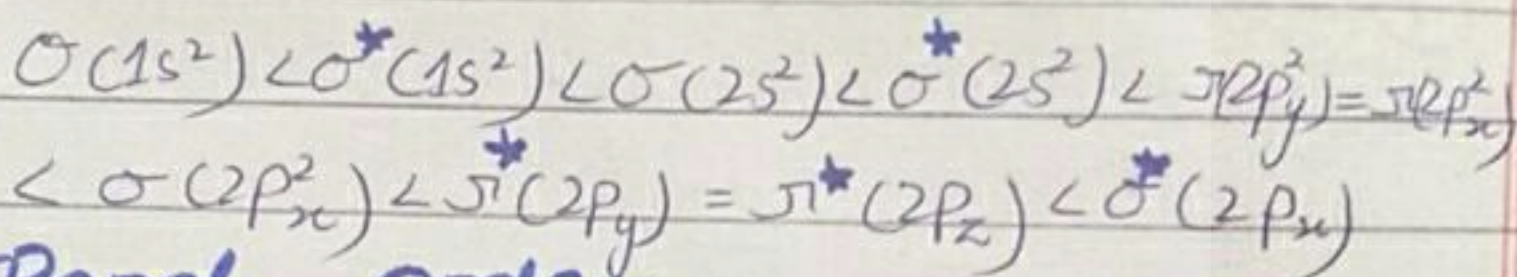
Molecular Orbital Energy level for N_2 molecule:-

Nitrogen (N_2):-

Electronic Configuration of N_2 is.



Energetics order of MO:-



Bond Order:-

$$= \frac{6 - 0}{2} = \frac{6}{2} = 3$$

From the electronic configuration of N_2 , it is clear that six electrons enter into three outermost bonding orbitals while no electron enters into anti-bonding orbitals. Thus the bond order in N_2 is 3, which corresponds to the triple bond consisting of one sigma and two π -bonds.

Bond Energy:-

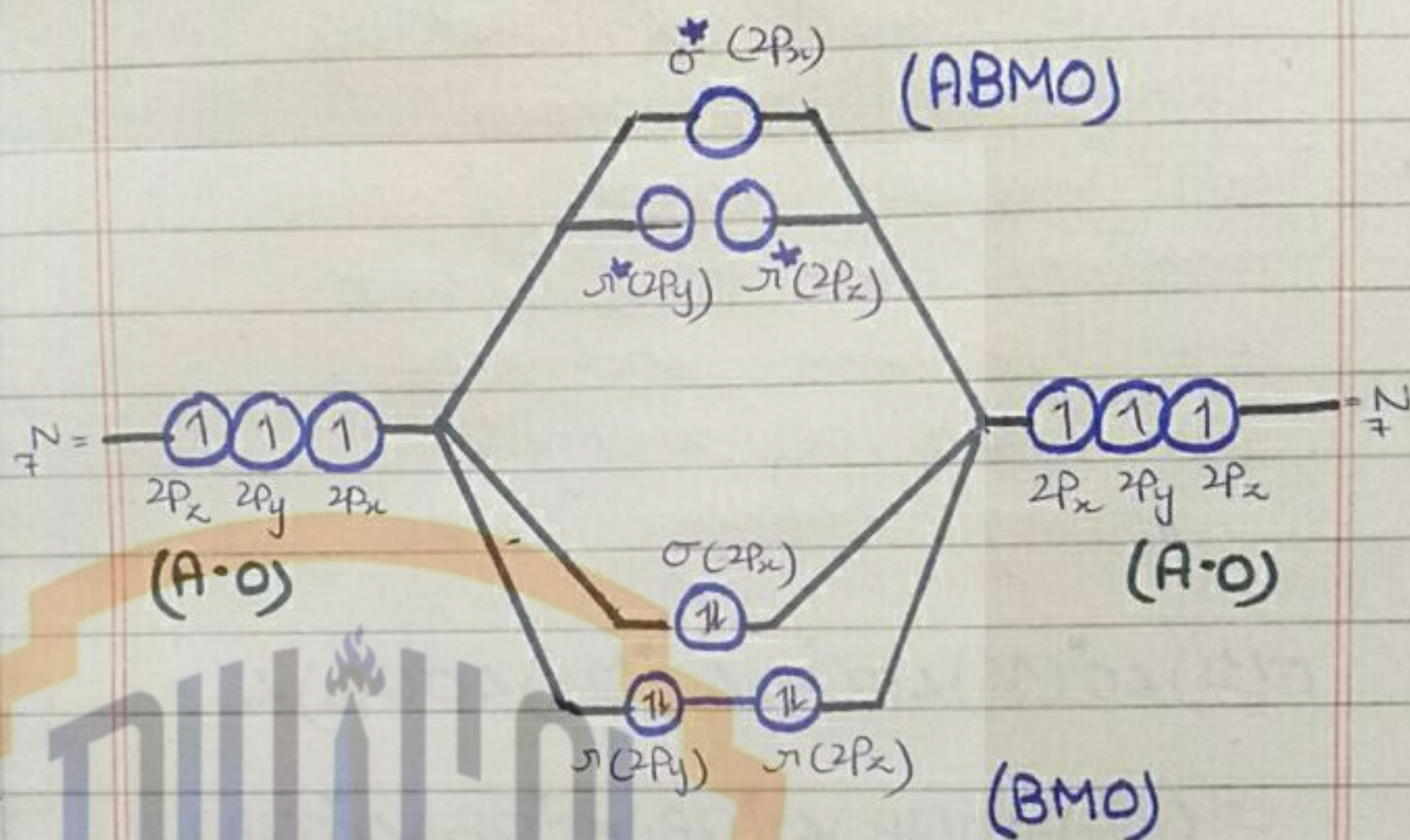
The bond energy of N_2 is very high = 941 kJ mol^{-1}

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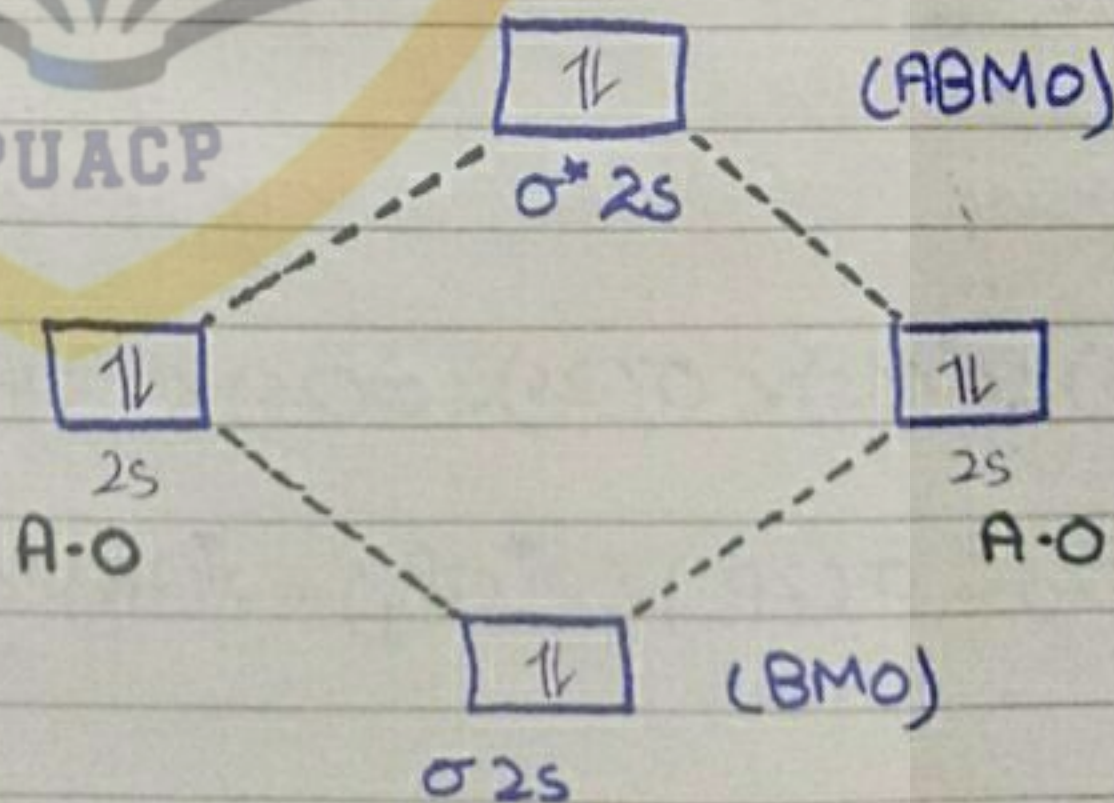
Diamagnetic property:-

Since no unpaired electrons are present in MOs so N_2 molecule show diamagnetic behaviour.



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Relative Energies of Molecular Orbitals:-

The energies of the molecular orbitals are determined by spectroscopic measurements.

⇒ The molecular orbitals of diatomic molecules such as O_2 , F_2 and their positive and negative ions can be arranged in following increasing order:-

$$\sigma(1s) < \sigma^*(1s) < \sigma(2s) < \sigma^*(2s) < \sigma(2p_x) <$$

$$\pi(2p_y) = \pi(2p_z) < \pi^*(2p_y) = \pi^*(2p_z) < \sigma^*(2p_x)$$

⇒ The diatomic molecules such as N_2 and other lighter molecules like B_2 , C_2 show slightly different energy order.

$$\sigma(1s) < \sigma^*(1s) < \sigma(2s) < \sigma^*(2s) < \pi(2p_y) =$$

$$\pi(2p_z) < \sigma(2p_x) < \pi^*(2p_y) = \pi^*(2p_z) < \sigma^*(2p_x)$$